§ 14 Quasiclassical Approximation

1. QC Expansion.

$$\psi(x) = e^{\frac{i}{\hbar}S} = e^{\frac{i}{\hbar}(S_0 + \hbar S_1 + \hbar^2 S_2 + \cdots)}$$

Schrodinger Equation.

$$\frac{1}{2m}(\nabla S)^2 + \frac{\hbar \nabla^2 S}{2m i} = E - U$$

(1) $\hbar \to 0 \Rightarrow$ H-J Eq: $\frac{1}{2m}(\nabla S_0)^2 = E - U$

$$\Rightarrow \begin{cases} p = \sqrt{2m(E-U)} \\ S_0 = \int p\,dx \end{cases}$$

QC condition : $|\hbar \nabla^2 S| \ll (\nabla S)^2$, $|\hbar \nabla p| \ll p^2$ fail @ $p = 0$

(2) Order of $\hbar$

$$\frac{1}{2m} 2\hbar (\nabla S_0)(\nabla S_1) + \frac{\hbar \nabla^2 S_0}{2mi} = E - U$$

$$\nabla S_1 = \frac{i}{2} \frac{\nabla p}{p} \Rightarrow S_1 = \frac{i}{2} \ln p \qquad e^{\frac{i}{\hbar} S_1} = \sqrt{\frac{1}{p}}$$

(a) $E > V$

$$\psi(x) = \frac{C}{\sqrt{p}} e^{\frac{i}{\hbar}\int p\,dx} + \frac{C}{\sqrt{p}} e^{-\frac{i}{\hbar}\int p\,dx}, \quad p = \sqrt{2m(E-U)}$$

(b) $E < V$

$$\psi(x) = \frac{D}{\sqrt{p}} e^{-\frac{1}{\hbar}\int p\,dx}, \quad p = \sqrt{2m(V-E)}$$



Connect 3 QC Region.

Near turning point → Linearize.

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + fx\psi = E\psi$$

$$\frac{d^2\psi}{dx^2} - \frac{2mf}{\hbar^2}\left(x - \frac{E}{f}\right)\psi = 0 \qquad x_0 - \frac{E}{f} = x - a = \alpha\xi$$

$$\frac{1}{\alpha^2}\frac{d^2\psi}{d\xi^2} - \frac{2mf}{\hbar^2}\alpha\xi\psi = 0$$

$$\psi''(\xi) - \xi\psi(\xi) = 0$$

Airy Function :

$$\psi(z) = \begin{cases} Ai(z) = \frac{1}{\pi}\sqrt{\frac{x}{3}}\, K_{1/3}\left(\frac{2}{3}x^{3/2}\right) \\ Bi(z) = \sqrt{\frac{x}{3}}\left[I_{1/3}\left(\frac{2}{3}x^{3/2}\right) + I_{-1/3}\left(\frac{2}{3}x^{3/2}\right)\right] \end{cases}$$

For Ai :

$$Ai(z \to \infty) \simeq \frac{1}{2\sqrt{\pi}}\, z^{-1/4}\, e^{-\frac{2}{3}z^{3/2}}$$

$$Ai(z \to -\infty) \simeq \frac{1}{\sqrt{\pi}}\, |z|^{-1/4} \sin\left(\frac{2}{3}|z|^{3/2} + \frac{\pi}{4}\right)$$

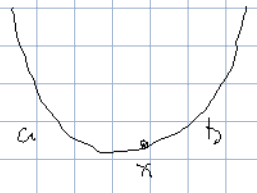
Analytic Continuation

$z = z e^{i\phi}$, $\phi = 0 \to \pi$ / $0 \to -\pi$

Quantization:

$\psi(-a) = 0$

$\Rightarrow -\frac{1}{\hbar}\int_{-a}^{0} p\,dx + \frac{\pi}{4} = n\pi \quad \Rightarrow \quad \int_{-a}^{0} p\,dx = \left(n-\frac{1}{4}\right)\pi\hbar$



WKB:

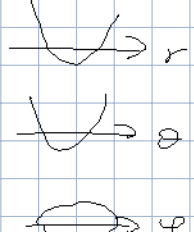
$$\frac{1}{\sqrt{p}}\sin\left(-\frac{1}{\hbar}\int_{b}^{x} p\,dx + \frac{\pi}{4}\right) = \frac{1}{\sqrt{p}}\sin\left(\frac{1}{\hbar}\int_{a}^{x} p\,dx + \frac{\pi}{4}\right)$$

$$= \frac{1}{\sqrt{p}}\sin\left(\frac{1}{\hbar}\int_{a}^{b} p\,dx + \frac{\pi}{2} - \frac{1}{\hbar}\int_{b}^{x} p\,dx - \frac{\pi}{4}\right)$$

$$\Rightarrow \quad \frac{1}{\hbar}\int_{a}^{b} p\,dx + \frac{\pi}{2} = n\pi$$

Example: H Atom

$E_z = -$



EKB :

$$p(r) = \nabla S(r)$$

$$\oint_{C_i} \vec{p} \cdot d\vec{q} = \oint \nabla S \cdot d\vec{q} = n_i h$$

Generalize ↓

$$= \left(n_i + \frac{m_i}{\bar{\varphi}}\right) h$$

$$\Rightarrow \frac{1}{2\pi} \oint p\,dx = \left[n + \frac{\tilde{m}}{\bar{\varphi}}\right) h$$

$$\oint p\,dx \sim nh$$

Quantum Tunneling.

